

Command Agent resolved conflicts between the input from sub agents. This agent decides whether to slow down or push the engines. It determines the final course and takes other decision.

Captain

Pilot

Officer of the Watch

This agent runs your pathfinding algorithms (like A* or DWA) and strictly adheres to maritime rules of the road (COLREGs) to avoid dynamic obstacles.

Helmsman –
Quarter Master

Actuator Agent receives a commanded heading and speed from the OOW and calculates the exact rudder angle and engine RPM needed to achieve it, accounting for the ship's physical momentum and drift (MOOS)

Lookout –
Able Seaman

Perception/Sensor Fusion Agent. It constantly ingests raw AIS, radar, and lidar data, classifies objects (e.g., "Fishing vessel, 2 miles, closing"), and passes only parsed, relevant tracks up to the Officer of the Watch

Quarter Master of the Watch - QMQ

The **Localization/State Agent.** It constantly cross-references GPS, inertial navigation, and depth soundings to confirm the ship's exact position on the digital chart (OpenSea.org)

Communi-
cations

VHF

Maintains the secure data links (like Link 16 or satellite comms)

Flag
signals

Uses optical cameras to read signal flags or Morse code flashes from the receiving warship

Chief Engineer

Propulsion

Electro-
Technical

Diagnostics

Cargo First Mate

Deck Safety
(Bosun)

Rig

Pumps

Executive Officer - XO

= Focus